**Dr. K. Prabakaran**

**Assistant Professor and Head i/c**

## Contact


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|---|---|
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## Academic Qualifications: M.Sc., Ph.D.

## Teaching Experience: 12 Years

## Research Experience: 16 Years

## Additional Responsibilities

1. Member - Board of Studies
2. SWAYAM Coordinator – Dept. Level
3. SWACHH Bharat coordinator – Dept. Level
4. NIRF Coordinator – Dept. Level
5. IQAC Coordinator – Dept. Level

## Areas of Research

**Geology - Hydrogeology, Remote sensing and GIS**

## Research Supervision / Guidance

| Program of Study | | Completed | Ongoing |
|---|---|---|---|
| Research | Ph.D. | - | 06 |
| | M.Phil. | 03 | 01 |
| Project | PG | 60 | 10 |
| | UG / Others | 15 | - |

## Publications

| International | | National | | Others |
|---|---|---|---|---|
| Journals | Conferences | Journals | Conferences | Books / Chapters / Monographs / Manuals |
| 16 | 04 | - | 04 | 02 |

**Cumulative Impact Factor (as per JCR) : 19.33**
**h-index : 05**
**Total Citations : 78**
**i10-index : 02**

## Funded Research Projects

**Completed Projects**

| S. No | Agency | Period | | Project Title | Budget (Rs. In lakhs) |
|---|---|---|---|---|---|
| | | From | To | | |
| 1 | AURF | 2016 | 2018 | "Studies on Mineral Resources and Coastal Geomorphology of Rameshwaram Region by using Remote Sensing and GISTechniques" | 0.80 |

## Events organized in leading roles

**Number of Seminars / Conferences / Workshops / Events organized:**

1. **Special Lecture** on "Employment opportunities for geology students" on 11-04-2023 in Department of Geology, Alagappa University, Karaikudi. (Organizing Secretary)

2. **Special Lecture** on "Fundamentals of Geological Mapping" association with GSI-Bhuvisamvad Activity on 22-02-2021in Department of Geology, Alagappa University, Karaikudi. (Organizing Secretary)

3. **Webinar Series** on "Petrological Studies" 18-05-2020, 27 & 28-05-2020 through online under RUSA Phase 2.0 sponsored in Department of Geology, Alagappa Univerity, Karaikudi. (OrganizingSecretary).

4. **National workshop on** "Late Quaternary Palaeo climatic and Evolving Hypothesis**"** on 07-01-2019 in Department of Geology, Alagappa Univerity, Karaikudi. (OrganizingSecretary).

5. **Special Lecture** on "Recent Advances in Petrology & Recent Advances in Geochemistry" from 19-12-2018 to 20-12-2018 in Department of Geology, Alagappa University, Karaikudi. (Organizing Secretary)

6. **International workshop on** "Recent trends in Biosciences (ICRTB 2016)**"** from 07- 04-2016 to 09-04-2016 in Alagappa Univerity, Karaikudi. (OrganizingSecretary).

7. **National workshop** on "Geospatial Technology in Coastal Studies" on 12-02-2016 and 13-02-2016 in Department of Oceanography and Coastal Area Studies, Alagappa Univerity, Thondi Campus. (OrganizingSecretary).

8. **DST** sponsored **Training Program** "**Geospatial technology – Theory and Practice using Open Source GIS Software"** on 26-08-2015 to 28-02-2015 in Department of Civil Engineering, Kalasalingam Univerisyt, Krishnankoil. (OrganizingSecretary).

9. **Faculty Development Program** "**Recent Innovations in Geotechnical Engineering**" on 05-05-2015 to 10-05-2015 in Department of Civil Engineering, Kalasalingam Univerisyt, Krishnankoil. (OrganizingSecretary).

10. **National level Technical Symposium "ASTHIVAAR-2012"** on 06-10-2012 in Department of Civil Engineering, Kalasalingam Univerisyt, Krishnankoil. (Organizing Secretary).

## Events Participated

**Conferences / Seminars / Workshops:**

1. Participated in the world water day – 2022 Webinar on "**Groundwater: making the invisible visible**" organized by the Department of Earth Sciences, **Pondicherry University**. Dated: 22-03-2022.

2. Participated in the Webinar on "**Water Chemistry Interpretation Methods Part-I**" organized by the Department of Geology, **Government Arts College, Salem**, Tamilnadu. Dated: 26-06-2020.

3. Participated in the Webinar on "**Basic Author Workshop Research Article Writing & Reference Management Using Mendeley**" organized by the Faculty of Science, **Annamalai University in collaboration with ELSEVIER**, Chidambaram, Tamilnadu, Dated: 22-06-2020.

4. Participated in the one-day National Workshop on "**Detailed petrology through innovative modern techniques in thin section preparation and petrographic studies**" organized by the PG & Research Department of Geology in association with **Chennai Metco, National College**, Tiruchirappalli, Tamilnadu. Dated: 20-02-2020.

5. Completed the self-learning Online Course on "**Understanding open Educational Resources**" offered by the **Common wealth of Learning, Canada**. Dated: 17-02-2020.

6. Coordinated in the Swachhta Hi Seva, SwachhtaPakhwads, RashtriyaPoshanMaah& Jal Shakti Abhiyan programmes Anti-plastic campaign rally at villages, organized by Centre for **SWACHH BHARAT** and **SWASTH BHARAT** (with financial support of **RUSA 2.0), Alagappa University**, Karaikudi from 01-07-2019 to 30-11-2019.

7. Participated in the Three-day International Workshop on "**E-Content Development for MOOCs among the Faculty Members**" organized by the Digital Education Cell, Sponsored by RUSA Phase 2.0 Scheme, **Alagappa University**, Karaikudi, Tamilnadu. Dated: 10th, 12th& 13-09-2019.

8. Participated in the one-day Seminar on "**Multidimensional Values of History for career development**" organized by the Department of History, **Alagappa University**, Karaikudi, Tamilnadu. Dated: 14-03-2019.

9. Participated in the one-day Workshop on "**Technical and Scholarly Writing" organized by the Technical and Academic writing centre**, Sponsored by RUSA Phase 2.0 Scheme, **Alagappa University**, Karaikudi, Tamilnadu.Dated:14-02-2019.

10. Coordinated the Cultural Competitions of Alagappa University Inter Collegiate **Talent Exhibit show-2019**, organized by Cultural Club, **Alagappa University**, Karaikudi from 02-02-2019 to 03-02-2019.

11. Coordinated in the **SWACHHTA HI SEVA 2018** Activity of Door-to-door meeting for sanitation behaviours, organized by Centre on SWACHH BHARAT and SWASTH BHARAT, **Alagappa University**, Karaikudi from 15-09-2018 to 02-10-2018.

12. Participated in the 100th**Orientation Programme** from 15-11-2017 to 12-12-2017 and obtained the **grade 'A'** organized by the **UGC-HRDC**, Bharathidasan University, Tiruchirappalli-620023, TamilNadu.

13. Participated in the DST Sponsored Two Days Workshop on "**Art of Project Writing and Management in Geospatial Technology**" Conducted by the DST Geospatial Chair, Centre for Remote Sensing, Bharathidasan University, Tiruchirappalli. Dated: 27-01-2017 & 28-01-2017

14. Participated in the National Training Workshop on "**LiDAR Technology and Applications**" organized by the Department of Earth Sciences, Annamalai University, Tamilnadu, Dated: 03-11-2016 &04-11-2016.

15. Participated in the International conference on **"Multi-dimensional view of Tamil People Culture"** organized by Alagappa University Tamil Panpaattu Centre, Karaikudi, Tamilnadu, Dated: 22-10-2016 &23-10-2016.

16. Participated in the International English Entry Level Examination "**Cambridge English Language Assessment Test"** (Entry 2, Business) and Achieved Council of Europe Level A2. Dated:13-07-2015.

17. Participated in the International Seminar on "**Smart city planning using Geographical Information Systems**" organized by Dept. of Civil Engineering, PSNA College of Engineering and Tecchnology, Dindigul. Dated10-12-2014.

18. Participated in the National Workshop on "**R&D Proposals to National and International Funding for Researchers - Approach & Preparation**" organized by Dept. of Chemical Engineering, National Institute of Technology, Tiruchirappalli. Dated 16-08-13 &17-08-13.

19. Participated in the National Technical Symposium on "**ASTHIVAR -2012**" organized by Dept. of Civil Engineering, Kalasalingam University, Krishnankoil. Dated 06-10- 2012.

20. Participated in the **National Level Conference** on "**NCODMM -2012**" organized by Dept. of Civil Engineering, Kalasalingam University, Krishnankoil. Dated 06 & 07-04- 2012.

21. Participated in the National Workshop on"**Frontiersin Catalysis Research**" organized by Dept. of Chemistry, National College, Tiruchirappalli. Dated 01-10-10.

22. Participated in the National Workshop on "**Environmental Protection**" organized by PWD (WRO) state Government and Dept. of Geology, National College, Tiruchirappalli. Dated24-09-10.

23. Participated in the National Workshop on "**Recent Developments in Geospatial Technology**" organized by Dept. of Geology, School of Geosciences, Bharathidasan University, Tiruchirappalli. Dated 18-03-09 to19-03-09.

24. Participated in the DST sponsored National Workshop on "**Coastal Non-Living Resources**" organized by Dept. of Geology, National College, Tiruchirappalli. Dated 13-03-08 to15-03-08.

25. Participated in the UGC sponsored National Seminar on "**Coastal Zones of India**" organized by Dept. of Geology, National College, Tiruchirappalli. Dated 06-12-07 to 07-12-07.

26. Participated in the National Workshop on "**Remote Sensing for Environmental Studies**" organized by Dept. of Geography, School of Geosciences, Bharathidasan University, Tiruchirappalli. Dated 21-02-05 to25-02-05.

## Other Training Programs

1. Participated online **Training** on **"**Wetland**s** in cities-spatial planning for disaster risk reduction**"** organized by **National Institute of Disaster Management,**

**Ministry of Home Affairs, Govt. of India** in collaboration with EP Department SPA, Bhobal. Dated 27-09-2022 & 29-09-2022.

2. Participated online **Training** on **"**Geospatial Technologies for Disaster Management**"** organized by **National Institute of Disaster Management, Ministry of Home Affairs, Govt. of India** in collaboration with Bharathidasan University, Tiruchirappalli. Dated 23-08-2022 & 25-08-2022.

3. Completed the **Online Refresher Course** on "Climate change: A guide for teachers of all disciplines" organized by **SWAYAM-ARPIT, IISER, Pune** (**UGC, Govt of India & Ministry of Education**). Dated: 01-12-2020 to 31-03-2021, Exam Date: 21-08-2021, Certificate Issued Date: 27-10-2021.

4. Participated in the **Two-Week Online** "Capacity Building programme for Faculty Members and Research Scholars" organized by the Alagappa University under the sponsorship of **UGC STRIDE Component-I Scheme**, Karaikudi, Tamilnadu. Dated: 12-06-2020 to 23-06-2020.

5. Participated in the Online Three-day National Level **Training Programme** on "**Coastal Disaster Risk Reduction & Resilience**" organized by National Institute of Disaster management, **Ministry of Home Affairs, New Delhi** in Collaboration with **Andhra Pradesh Human Resource Development Institute**, Bapatla, Guntur District. Dated: 17-06-2020 to 19-06-2020.

6. Completed the **Online Refresher Course** on "Curriculum pedagogy and evaluation for higher education" organized by **SWAYAM-ARPIT**, Central University of Kerala (**UGC, Govt of India, Ministry of Human Resource Development**). Dated: 01-09-2019 to 15-01-2020, Exam Date: 15-09-2020, Certificate Issued Date: 20-01-2021

7. Participated Short term **Training programme** on **"Training and scope of SCUBA Diving"** organized by Department of Oceanography and CAS, Thondi Campus, Alagappa University. Dated 15-02-2017 &16-02-2017.

8. Participated in the **Training programme** on "**Open-source Quantum GIS**" organized by Dept. of Civil Engineering, Velammal College of Engineering and Technology, Madurai. Dated23-01-2013.

9. Participated ISRO, Govt. of India sponsored **Training programme**on "**2nd course on Geoinformatics for Disaster Management**" organized by **Geological Survey of India Training Institute, Hydrabad**. Dated 11-07-06 to09-08-06.

10. Participated in the **Training programme** on "**Introduction to Arc GIS 9**"organized by **ESRI India, NIIT GIS Ltd**. Chennai. Dated 27-02-06 to 03-03-06.

11. Participated in the **ISRO – NNRMS Training programme** on "**Geological Remote Sensing and GIS**". Dept. of Geology, **Anna University**, Chennai. Dated 5-12-05 to 23-12-05.

12. Participated in the **Training programme** on "**Mathematical Modeling of Ground Water Flow and Mass transport through theory and hands on**

**training**" organized by School of Environmental Sciences, **Jawaharlal Nehru University,** New Delhi. Dated 12-11-04 to09-12-04.

13. Onboard participated on **C.R.V. Sagar Paschimi for the research programme** sponsored by **NIOT**, Dept. of Ocean Development, Govt. of India. Dated15-09-03 to 18-09-03.

## Resource persons in various capacities

**Number of Invited / Special Lectures delivered:**

1. Act as a **Jury Member** on "one day workshop on internal Hackathon and reward best ideas of YUKTI-NIR-2023" in the Entrepreneurship, innovation and career hub in association with institution innovation council, Alagappa University, Karaikudi. Dated: 24-02-2023.

2. Act as a **Co-Chair person** on the session topic "Volcanism and meteorite impacts: Global impacts: Global climatic changes and pollutin" in the National Seminar on "Emerging Pollutant in Geosphere: Impact on Environment, Management and Challenges" organized by Department of Geology, Malankara Catholic College, Mariagiri, Kanyakumari. Dated: 9 & 10-07-2021.

3. Delivered a Lecture on "**Disaster Management**" organized by Dept. of Civil Engineering, Anjalai Ammal - Mahalingam Engineering College, Koyilvenni, Thiruvarur District, Tamilnadu. Dated:10-09-2019.

4. Delivered a Special Lecture on **"Remote Sensing and Geographic Information System (GIS) in Civil Engineering Application**" organized by Department of Civil Engineering, Alagappa Polytechnic college, Karaikudi, Tamilnadu. Dated:16-08-2018.

5. Delivered a Lecture in National Seminar **"GIS and Remote Sensing Technology for National Development**" organized by Department of Civil Engineering, Kalasalingam Univerisyt, Krishnankoil, Tamilnadu. Dated:10-10-2014.

6. Delivered a Lecture in National Seminar on "**Recent Trends in Civil Engineering**" organized by Dept. of Civil Engineering, Mountzion College of Engineering and Technology, Puthukottai, Tamilnadu. Dated:16-08-2012.

7. Delivered a Lecture in National Seminar on "**Remotesensing and GIS**" organized by Dept. of Civil Engineering, Kalasalingam University, Krishnankoil, Tamilnadu, Dated: 25 &26-02-2012.

## Paper Presented

1. Presented paper entitled "Occurrence, spatial distribution and characterization of microplastics from the intertidal beach sediments in the Gulf of Mannar coast, Tamilnadu, India" N**ational Seminar** on Marine Pollution and Ecological degradation MPAED-2022, Dept. of Geology, Malankara Catholic College, Mariagiri, Kanyakumari-629153. Dated:27.07.2022 to29.07.2022.

2. Presented paper entitled "Exploration on Mineral resources and coastal geomorphology of Rameshwaram Island by using Remotesensing and GIS Techniques" **International Conference** on Geological and Environmental Sustainability (ICGES), SCIENCEFORA, held in Putrajaya, **Malaysia** on 03-09-2019.

3. Presented paper entitled "Signatures of Climate change in velar river delta region, Puddukkottai district, Tamilnadu, India" **International Conference** on Impact of Climate change on Water resources, Centre for Geoinformatics and Planetary Studies, **Periyar University**, Salem-636011. Dated:12.07.2018 to13.07.2018.

4. Presented paper entitled "Sustainable open cast mining at small mines" **National Seminar** on "Twenty Ninth National Convention of Mining Engineers" held in Institution of Engineers (India), Salem local centre in association with the Dept. of Geology, **Periyar University and Geo Exploration & Mining Solutions** (GEMS), Salem-636011. from 22-02-2018 to 23-02-2018.

## Publications

**1.** Muruganantham Arumugam, **Prabakaran Kulandaisamy**, Sivakumar Karthikeyan, Kongeswaran Thangaraj, Venkatramanan Senapathi, Sang Young Chung, Subagunasekar Muthuramalingam, Muthuramalingam Rajendran, Sathish Sugumaran and Siva Manimuthu. **(2023)** "An Assessment of Geospatial analysis combined with AHP Techniques to Identify Groundwater Potential Zones in the Pudukkottai District, Tamil Nadu, India.", **Water** (MDPI), 15, (1101)pp. 1-21. **Impact Factor: 3.530**

**2.** Muruganantham Arumugam, Sivakumar Karthikeyan, **Prabakaran Kulandaisamy**, Kongeswaran Thangaraj, Venkatramanan Senapathi, Bangaru Priyanga Sundaram, Agastheeswaran Vellaikannu and Perumal Velmayil (**2022**) "Role of GIS in deciphering hydrogeochemical processes and quality in Pudukottai district, Tamil Nadu, India", Environmental Earth Sciences (**springer**), Vol. 81(18) pp.1-22(445)**. Impact Factor: 3.119.**

3. Sivakumar Karthikeyan, **Prabakaran Kulandaisamy**, Venkatramanan Senapathi, Sang Yong Chung, Kongeswaran Thangaraj, Muruganantham Arumugam, Sathish Sugumaran, and Sung Ho-Na **(2022)** "Hydrogeochemical Survey along the Northern Coastal Region of Ramanathapuram District, Tamilnadu, India." Applied Sciences, Vol. 12(11), pp.1-28 (5595), **Impact Factor:2.838.**

4. Sivakumar Karthikeyan, **Prabakaran Kulandaisamy**, Muthusamy S, Kongeswaran Thangaraj, Muruganantham A, Gnanachandrasamy G. (**2022**) "Agriculture Drought Management in Ramanathapuram District of Tamil Nadu, India." Journal of Climate Change, (**springer**), Vol 8, (1), pp59-65. **Impact Factor: 4.743.**

5. Balaji Lakshmana Rao, Muthukannan Muthiah, Naveen Raj Thangasamy, **Prabakaran Kulandaisamy** (**2021**) "Advent Geospatial Technologies in Land Evaluation: A Critical Review" International Journal of Engineering Trends and Technology, Vol.69, Issue 8, pp.190-205.

6. Muruganantham Arumugam, Sivakumar Karthikeyan, Kongeswaran Thangaraj, **Prabakaran Kulandaisamy**, Bangaru Priyanga Sundaram, Karikalan Ramasamy, Agastheeswaran Vellaikannu and Perumal Velmayil (**2021**) "Hydrogeochemical analysis for Groundwater suitability appraisal in Sivagangai, Economically Backward District of Tamilnadu", Journal Geological Society of India (**springer**), Vol.97, pp.789-798.**Impact Factor: 1.613.**

7. Sivakumar Karthikeyan, Shanmugasundaram Arumugam, Jayaprakash Muthumanickam, **Prabakaran Kulandaisamy**, Muthusamy Subramanian, Ramachandran Annadurai, VenkatramananSenapathi, Selvam Sekar**(2021)** "Causes of heavy metal contamination in groundwater of Tuticorin industrial block, Tamil Nadu, India" Environmental Science and Pollution Research (**springer**), Vol.28, pp.18651-18666.**Impact Factor: 4.223**

8. **Prabakaran Kulandaisamy**, Sivakumar Karthikeyan &ArunaChockalingam, **(2020)** "Use of GIS-AHP tools for potable groundwater potential zone investigations—a case study in Vairavanpatti rural area, Tamil Nadu, India" Arabian Journal of Geosciences(**springer**), Vol.13 (17) Article number: 866 pp.1-15. **Impact Factor: 1.827.**

9. Naveen Raj T, **Prabakaran K**, Udhaya Gowtham G, Mahesh Kumar M, Aravind D, Saravana Vikash A (2016) "Landslide Hazard zonation mapping using Geospatial Technology in Bodimettu hills, Theni district, Tamilnadu" **International Journal of Earth Sciences and Engineering**, Vol.9 (4), pp.1454-1457.

10. Kumar.G, **Prabakaran. K** and Selvam.G (2015) "Assessment of Groundwater Quality for Veppanthattai Taluk, Perambalur District, Tamil Nadu Using Remote Sensing and GIS Techniques" International Journal of Recent Scientific Research, Vol. 6, Issue 3, pp 3142-3146.

11. Selvam. G, **Prabakaran. K**, Srinivasan. D and Kumar. G, (2015) "Assessmentof Groundwater Quality in And Around Manapparai Block, Tamilnadu (India), International Journal of Recent Scientific Research, Vol. 6, Issue 3, pp 2903-2907.

12. Sivapragasam. C, D. Venkat reddy, **Prabakaran. K**, M. Vigneswaran, S. Senthilkumaran, C.Sivaprasath and M. Varun kumar (2012) "Physico-Mechanical Properties of Select

Granitoidal Rocks from a Part of Pandiyan Mobile Belt, India", International Journal of Earth Sciences and Engineering, Vol.5 (3), pp437- 441.

13. **Prabakaran. K** and Anbarasu. K (2010) "Coastal Geomorphology and Evolution of Rameswaram Island, Tamilnadu, India". Research Journal of Earth Sciences, Vol.2 (2), pp30-35.

14. **Prabakaran. K** and Anbarasu. K (2010) "Evolution of Vaigai delta, Tamilnadu, India (East coast) during Quaternary". International Journal of Geomatics and Geosciences, Vol. 1(2), pp 211-222.

15. **Prabakaran. K** and Anbarasu. K (2010) "Evidences for late Quaternary Sea level changes along the coast of Tamilnadu, East coast of India". International Journal of Recent Scientific Research, Vol. 7, pp 163-167.

**Book Chapter**

1. **Book**: Issues of Coastal Groundwater Contamination**, "Chapter**- Groundwater Contamination in Coastal Aquifers" Sivakumar Karthikeyan, **Prabakaran Kulandaisamy**, Venkatramanan Senapathi, Kongeswaran Thangaraj, & Muruganantham Arumugam. **Publisher: Elsevier** (2022) pp.9-18.